

Problem 1

Consider a pump designed for incompressible liquids shown in the figure below. The size of the pump is characterized by the impeller diameter (d); it is assumed that all other dimensions of the pump scale with this single length scale parameter. A motor is connected to the pump that spins the impeller at a constant angular velocity (Ω). In the test setup shown in the figure, the pump pulls in liquid of density ρ from the reservoir at pressure P_1 , and the liquid discharges at the pump exit at pressure P_2 . A valve is used to adjust backpressure on the pump to simulate different flow resistances.

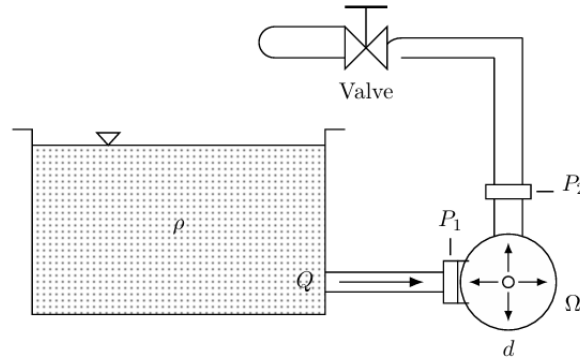


Table 1: Pump head ($H = \Delta P/(\rho g)$) generated as a function of flow rate (Q) at different pump speeds.

Speed = 1000 rpm		Speed = 1475 rpm		Speed = 1990 rpm	
Flowrate [gpm]	Head [ft]	Flowrate [gpm]	Head [ft]	Flowrate [gpm]	Head [ft]
0.0	30.4	0.0	64.7	0.0	113.4
23.5	29.7	12.6	63.6	9.3	113.1
38.7	29.0	33.8	62.4	22.7	111.0
51.0	28.6	65.1	60.3	36.5	110.9
74.8	27.0	96.4	57.3	59.9	110.1
87.8	25.2	115.8	56.0	81.9	109.0
101.2	24.3	139.9	53.8	107.5	105.0
120.6	22.8	149.6	54.2	126.9	105.2
		176.8	49.6	151.9	101.8
		188.0	47.2	173.4	100.1
				189.1	98.3
				208.1	95.9
				226.6	93.5
				240.0	90.6

Part (a)

Perform dimensional analysis following the steps presented in class to determine the non-dimensional relation between nondimensional pressure rise across the pump and the other variables. You can assume that the viscosity of the liquid can be neglected.

Let $\Delta P = P_2 - P_1$ be the pressure rise across the pump. Neglecting viscosity, we have:

$$\Delta P = f(\rho, Q, \Omega, d)$$

The fundamental dimensions of the variables are:

	ΔP	ρ	Q	Ω	d
M	1	1	0	0	0
L	-1	-3	3	0	1
τ	-2	0	-1	-1	0

With $N = 5$ variables and 3 fundamental dimensions (M, L, τ), the dimensional matrix has rank $R = 3$, giving $N - R = 2$ dimensionless groups. Choosing ρ, Ω, d as repeating variables:

For Π_1 , containing ΔP :

$$\begin{aligned}\Pi_1 &= \Delta P \rho^a \Omega^b d^c \\ M^0 L^0 \tau^0 &= (ML^{-1}\tau^{-2})(ML^{-3})^a(\tau^{-1})^b(L)^c\end{aligned}$$

Matching exponents:

$$\begin{aligned}M : 1 + a &= 0 \Rightarrow a = -1 \\ L : -1 - 3a + c &= 0 \Rightarrow c = -2 \\ \tau : -2 - b &= 0 \Rightarrow b = -2 \\ \Pi_1 &= \Delta P \rho^{-1} \Omega^{-2} d^{-2} = \frac{\Delta P}{\rho \Omega^2 d^2}\end{aligned}\tag{1}$$

For Π_2 , containing Q :

$$\begin{aligned}\Pi_2 &= Q \rho^a \Omega^b d^c \\ M : a &= 0 \\ L : 3 - 3a + c &= 0 \Rightarrow c = -3 \\ \tau : -1 - b &= 0 \Rightarrow b = -1 \\ \Pi_2 &= Q \Omega^{-1} d^{-3} = \frac{Q}{\Omega d^3}\end{aligned}\tag{2}$$

By the Buckingham Pi theorem, $\Pi_1 = f(\Pi_2)$:

$$\frac{\Delta P}{\rho \Omega^2 d^2} = f\left(\frac{Q}{\Omega d^3}\right)\tag{3}$$

Defining the pump head as $H \equiv \Delta P/(\rho g)$, this becomes

$$\boxed{\frac{gH}{\Omega^2 d^2} = f\left(\frac{Q}{\Omega d^3}\right)}\tag{4}$$

i.e. the head coefficient $\psi = gH/(\Omega^2 d^2)$ is a function of the flow coefficient $\phi = Q/(\Omega d^3)$. $\square$

Part (b)

Plot the data in the table above in non-dimensional form using the non-dimensional variables determined in your analysis.

The flow rate and head data at each of the three pump speeds (1000, 1475, and 1990 rpm) were converted to SI units ($\text{gpm} \rightarrow \text{m}^3/\text{s}$, $\text{ft} \rightarrow \text{m}$, $\text{rpm} \rightarrow \text{rad/s}$) and combined with $g = 9.81 \text{ m/s}^2$ to compute $\phi = Q/(\Omega d^3)$ and $\psi = gH/(\Omega^2 d^2)$. The impeller diameter d is not given in the problem; since the same physical pump ($d = \text{const}$) is used at all three speeds, an arbitrary but consistent value of d only rescales both axes uniformly and does not affect whether the three speed curves collapse onto a single curve, so $d = 1$ was used as a placeholder value. The resulting nondimensional performance curves are shown in Figure 1

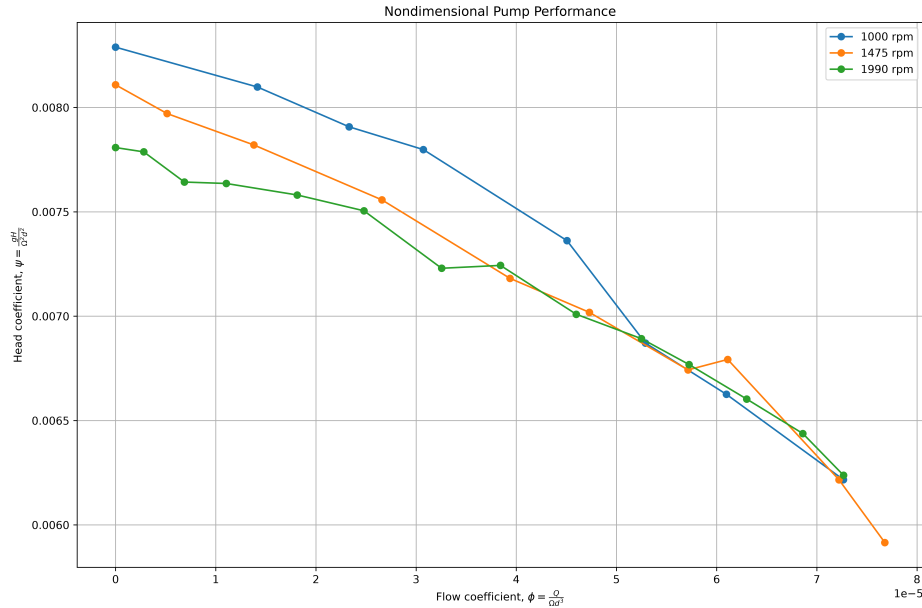


Figure 1: Nondimensional pump performance: head coefficient $\psi = gH/(\Omega^2 d^2)$ vs. flow coefficient $\phi = Q/(\Omega d^3)$ for three impeller speeds.

Part (c)

Does your analysis appear to have identified the correct non-dimensional relationship? Why?

Largely, yes. The three speed curves converge closely onto a single curve $\psi = f(\phi)$ across most of the flow range, namely at moderate-to-high flow coefficients, which means that ϕ and ψ are the dominant governing nondimensional groups for this pump. There is a small systematic spread between the curves near shutoff (small ϕ), where ψ decreases slightly with increasing speed rather than collapsing exactly. This residual is expected because part (a) neglected viscosity; whereas in the real physical pump, viscous and internal losses (captured by a Reynolds number $\text{Re} = \rho\Omega d^2/\mu$ which was assumed to be sufficiently negligible) are not perfectly similar across different speeds, since Reynolds number increases with Ω . So our inviscid analysis captures the correct primary relationship. $\square$

Problem 2

A gas of non-interacting particles of mass m at temperature T has density ρ , and internal energy per unit volume ϵ .

Part (a)

Using dimensional analysis, determine how ϵ must depend on ρ, T , and m . In your formulation use $k_B = \text{Boltzmann's constant}$, $h = \text{Planck's constant}$, and $c = \text{speed of light}$ to include possible quantum and relativistic effects.

We assume $\epsilon = f(\rho, T, m, k_B, h, c)$. The fundamental dimensions of the variables are:

	ϵ	ρ	T	m	k_B	h	c
M	1	1	0	1	1	1	0
L	-1	-3	0	0	2	2	1
τ	-2	0	0	0	-2	-1	-1
T	0	0	1	0	-1	0	0

With $N = 7$ variables and 4 fundamental dimensions (M, L, τ, T), the dimensional matrix has rank $R = 4$, giving $N - R = 3$ dimensionless groups. Choosing m, k_B, h , and c as repeating variables:

For Π_1 , containing ϵ :

$$\Pi_1 = \epsilon m^a k_B^b h^c c^d$$

$$M^0 L^0 \tau^0 T^0 = (ML^{-1}\tau^{-2})(M)^a (ML^2\tau^{-2}T^{-1})^b (ML^2\tau^{-1})^c (L\tau^{-1})^d$$

Matching exponents:

$$\begin{aligned} M : 1 + a + b + c &= 0 \\ L : -1 + 2b + 2c + d &= 0 \\ \tau : -2 - 2b - c - d &= 0 \\ T : -b &= 0 \end{aligned}$$

Solving this system gives $a = -4, b = 0, c = 3, d = -5$. Thus:

$$\Pi_1 = \epsilon m^{-4} h^3 c^{-5} = \frac{\epsilon h^3}{m^4 c^5} \quad (5)$$

For Π_2 , containing ρ , the identical procedure gives $a = -4, b = 0, c = 3, d = -3$:

$$\Pi_2 = \rho m^{-4} h^3 c^{-3} = \frac{\rho h^3}{m^4 c^3} \quad (6)$$

For Π_3 , containing T , the identical procedure gives $a = -1, b = 1, c = 0, d = -2$:

$$\Pi_3 = T m^{-1} k_B c^{-2} = \frac{k_B T}{m c^2} \quad (7)$$

By the Buckingham Pi Theorem, $\Pi_1 = f(\Pi_2, \Pi_3)$. Rearranging for ϵ gives:

$$\boxed{\epsilon = \frac{m^4 c^5}{h^3} f\left(\frac{\rho h^3}{m^4 c^3}, \frac{k_B T}{m c^2}\right)} \quad (8)$$

Part (b)

Consider the limit of slow moving particles without quantum effects by requiring c and h to drop out of your dimensionless formulation. How does ϵ depend on ρ and T ? What type of gas follows this thermodynamic law?

Assume a power-law form for the arbitrary function, $f(\Pi_2, \Pi_3) = \gamma (\Pi_2)^\alpha (\Pi_3)^\beta$, where $\gamma \in \mathbb{R}$. Expanding this into the relation for ϵ :

$$\epsilon = \gamma \frac{m^4 c^5}{h^3} \left(\frac{\rho h^3}{m^4 c^3} \right)^\alpha \left(\frac{k_B T}{m c^2} \right)^\beta = \gamma \rho^\alpha T^\beta k_B^\beta m^{(4-4\alpha-\beta)} h^{(3\alpha-3)} c^{(5-3\alpha-2\beta)}$$

To ensure h and c drop out, their exponents must equal zero:

$$h : 3\alpha - 3 = 0 \Rightarrow \alpha = 1$$

$$c : 5 - 3\alpha - 2\beta = 0 \Rightarrow \beta = 1$$

Substituting $\alpha = 1$ and $\beta = 1$ back into the expression gives:

$$\boxed{\epsilon = \gamma \frac{\rho k_B T}{m}} \quad (9)$$

The internal energy per unit volume depends linearly on density and temperature. This follows the thermodynamic law for an **ideal gas** (ideal gas law). $\square$

Part (c)

Consider the limit of massless particles (i.e. photons) by requiring m and ρ to drop out of your dimensionless formulation of part a). How does ϵ depend on T in this case? What is the name of this radiation law?

By the expanded power-law equation, we require the exponents of m and ρ to be zero:

$$\rho : \alpha = 0$$

$$m : 4 - 4\alpha - \beta = 0 \Rightarrow \beta = 4$$

Substituting $\alpha = 0$ and $\beta = 4$ back into the expression:

$$\boxed{\epsilon = \gamma \frac{k_B^4 T^4}{h^3 c^3}} \quad (10)$$

The internal energy per unit volume is proportional to T^4 . This is the **Stefan-Boltzmann Law**. $\square$

Problem 3

The vertical motion of a ship in water is governed by the z -component of the Navier–Stokes equation:

$$\frac{\partial w}{\partial t} + u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} = -\frac{1}{\rho} \frac{\partial p}{\partial z} - g + \frac{\mu}{\rho} \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right)$$

where u, v, w are the x, y , and z velocities respectively, and all other variables have their normal meanings. The flow velocity far from the ship is given by U and the length of the ship at the waterline is l .

Part (a)

Non-dimensionalize the governing equation, the z -component of Navier–Stokes.

We define characteristic scales to normalize the variables:

$$x^* = \frac{x}{l}, \quad y^* = \frac{y}{l}, \quad z^* = \frac{z}{l}, \quad u^* = \frac{u}{U}, \quad v^* = \frac{v}{U}, \quad w^* = \frac{w}{U}, \quad t^* = \frac{tU}{l}, \quad p^* = \frac{p}{\rho U^2}$$

Substituting these scales into each term of the governing equation (with $\nu = \mu/\rho$ the kinematic viscosity):

$$\begin{aligned} \partial_t w &= \frac{U^2}{l} \partial_{t^*} w^*, & u \partial_x w &= \frac{U^2}{l} u^* \partial_{x^*} w^*, \\ v \partial_y w &= \frac{U^2}{l} v^* \partial_{y^*} w^*, & w \partial_z w &= \frac{U^2}{l} w^* \partial_{z^*} w^*, \\ \frac{1}{\rho} \partial_z p &= \frac{U^2}{l} \partial_{z^*} p^*, & \frac{\mu}{\rho} \nabla^2 w &= \frac{\nu U}{l^2} \nabla^{*2} w^* \end{aligned}$$

Substituting these into the governing equation gives:

$$\frac{U^2}{l} (\partial_{t^*} w^* + u^* \partial_{x^*} w^* + v^* \partial_{y^*} w^* + w^* \partial_{z^*} w^*) = -\frac{U^2}{l} \partial_{z^*} p^* - g + \frac{\nu U}{l^2} \nabla^{*2} w^*$$

Dividing through by the convective acceleration term (U^2/l) gives:

$$\boxed{\partial_{t^*} w^* + u^* \partial_{x^*} w^* + v^* \partial_{y^*} w^* + w^* \partial_{z^*} w^* = -\partial_{z^*} p^* - \frac{gl}{U^2} + \frac{\mu}{\rho U l} \nabla^{*2} w^*} \quad (11)$$

Part (b)

What common non-dimensional parameters result?

The resulting common non-dimensional parameters are:

- Inverse Froude Number squared: $\frac{1}{Fr^2} = \frac{gl}{U^2}$
- Inverse Reynolds Number: $\frac{1}{Re} = \frac{\mu}{\rho U l}$

Part (c)

Assume you didn't have knowledge of the governing equation but knew that the z -velocity, w , was a function of ρ, g, μ, U , and l . Find the non-dimensional parameters using the Buckingham PI theorem.

We assume $w = f(\rho, g, \mu, U, l)$. The fundamental dimensions of the variables are:

	w	ρ	g	μ	U	l
M	0	1	0	1	0	0
L	1	-3	1	-1	1	1
τ	-1	0	-2	-1	-1	0

With 6 variables (w, ρ, g, μ, U, l) and 3 fundamental dimensions (M, L, τ), we choose ρ, U, l as repeating variables to form $6 - 3 = 3$ dimensionless groups.

For Π_1 , containing w :

$$\Pi_1 = w \rho^a U^b l^c$$

$$M : a = 0, \quad L : 1 - 3a + b + c = 0, \quad \tau : -1 - b = 0 \Rightarrow a = 0, \quad b = -1, \quad c = 0$$

$$\Pi_1 = \frac{w}{U} \quad (12)$$

For Π_2 , containing g : solving analogously gives $a = 0, \quad b = -2, \quad c = 1$:

$$\Pi_2 = \frac{gl}{U^2} \quad (\text{Inverse Froude Number squared}) \quad (13)$$

For Π_3 , containing μ : solving analogously gives $a = -1, \quad b = -1, \quad c = -1$:

$$\Pi_3 = \frac{\mu}{\rho U l} \quad (\text{Inverse Reynolds Number}) \quad (14)$$

By the Buckingham Pi theorem, $\Pi_1 = f(\Pi_2, \Pi_3)$, i.e.

$$\boxed{\frac{w}{U} = f(Fr, Re) = f\left(\frac{U}{\sqrt{gl}}, \frac{\rho U l}{\mu}\right)} \quad (15)$$

This matches the parameters found directly from the governing equation in part (b), as expected: pressure was not included among the given variables here, so no Euler-number-like group appears. $\square$

Part (d)

We wish to test a scaled model of a tanker ship at scale of 1/100 the actual size, is it practical to achieve dynamic similarity?

Let subscripts m and p denote the model and prototype (full-scale), respectively, with $l_m/l_p = 1/100$. Dynamic similarity requires matching both the Froude number and the Reynolds number:

$$Fr_m = Fr_p \Rightarrow \frac{U_m}{\sqrt{gl_m}} = \frac{U_p}{\sqrt{gl_p}} \Rightarrow \frac{U_m}{U_p} = \sqrt{\frac{l_m}{l_p}} = \frac{1}{10}$$

$$Re_m = Re_p \Rightarrow \frac{U_m l_m}{\nu_m} = \frac{U_p l_p}{\nu_p} \Rightarrow \frac{\nu_m}{\nu_p} = \left(\frac{U_m}{U_p}\right) \left(\frac{l_m}{l_p}\right) = \left(\frac{1}{10}\right) \left(\frac{1}{100}\right) = \frac{1}{1000}$$

So Froude scaling requires the model velocity to be 1/10 of the prototype velocity, and simultaneously satisfying Reynolds scaling in this case requires a model fluid with kinematic viscosity 1/1000 that of the prototype fluid. Taking $\nu_p \approx 1 \times 10^{-6} \text{ m}^2/\text{s}$ (water at 20°C) gives $\nu_m \approx 1 \times 10^{-9} \text{ m}^2/\text{s}$, roughly 1000× less viscous than water. No practical fluid has a kinematic viscosity anywhere near this low.

It is therefore **not practical** to achieve full dynamic similarity: matching the Froude number and the Reynolds number simultaneously would require a model test fluid whose kinematic viscosity is unrealistically small, so the two scaling requirements cannot both be satisfied with any real fluid. $\square$